

Masking cannot remove what is inside the region of interest: location-dependent shortcut mitigation across dermoscopy, ultrasound and capsule endoscopy

Abstract

Medical image classifiers exploit acquisition artifacts—hair, sonographer calipers, intestinal debris—that correlate with the diagnosis; the standard defence restricts the model to the region of interest (ROI). We show that this defence fails for artifacts inside the ROI, give a closed-form account of when and why, and evaluate alternatives. In pre-registered, leakage-controlled traps built from real artifacts in four diseases, masking always helps more for out-of-ROI than for in-ROI artifacts (crossover +0.15 to +0.37 reversed-test AUROC, all Holm-significant), with frozen and fine-tuned backbones and in the zero-shot scores of vision–language foundation models. Masking becomes harmful when it also removes disease context: in dermoscopy, controlled overlap sweeps, fine-tuning (capsule -0.28) and unaltered test data (official thyroid split -0.10 on shortcut-conflicting cases; two of three held-out dermoscopy hospitals). A linear-Gaussian model fitted only to each classifier’s clean and correlated performance matches its held-out reversed-test AUROC (mean absolute error 0.039) and the location crossover across 14 cohort–backbone pairs ($r = 0.997$). For frozen foundation-model features we propose mask-then-erase with generic synthetic overlays (U-MtE), which needs no artifact examples, artifact masks or artifact labels—only the ROI mask. It removes in-ROI shortcuts carried by distinct overlays (thyroid +0.13 to +0.22 over masking on three backbones); erasure beats augmentation with the same overlays (thyroid, dermoscopy); a disease-protected variant prevents failure on pathology-like artifacts; text-prompted erasure is a near no-op. Shortcuts carried by correlates of the artifact (hair, chest drains; chest radiography is a boundary case) require label-based reweighting. Code, pre-registrations and a backbone-agnostic tool are public.

Keywords: shortcut learning, spurious correlation, artifacts, dermoscopy, chest radiography, concept erasure

1. Introduction

Deep classifiers for medical images frequently rely on features that are correlated with the label in the training distribution but are not evidence of disease [9, 7]. In dermoscopy, surgical skin markings [29] and ruler scale bars [30] changed the melanoma predictions of a market-approved system; in chest radiography, pneumothorax classifiers detect the chest drain that treats the condition rather than the condition itself

[19, 12, 32]. Such correlations are causal in origin—suspicious lesions are measured and marked, collapsed lungs are drained—so they are present in every retrospectively collected dataset.

The most widely recommended defence is to restrict the classifier to the region of interest (ROI): segment the lesion or the lungs and discard the rest of the image. The rationale assumes that the shortcut lives in the background. Rulers, hair, ink and drains do not respect that assumption. When the artifact overlaps the ROI, masking cannot remove it; it removes everything *else*. We ask whether the effect of a mitigation depends on where the artifact sits, which quantity governs that dependence, what supervision each family of mitigations actually needs, and whether the in-ROI regime can be mitigated without localising the artifact.

Contributions.

1. **A location law for ROI masking**, established with pre-registered real-artifact traps in four diseases and three imaging modalities, replicated with three frozen backbone families and three model sizes on real artifacts (DINOv2 ViT-S/B/L, MedSigLIP, ConvNeXt; the dermatology model DermLIP shows the reversal in controlled sweeps but harm at both hair locations) and with end-to-end fine-tuning (ResNet-50, ViT-S), confirmed on an unaltered test distribution, and robust to the artifact-label definition (Secs. 7–8).
2. **Foundation models**: the location effect appears in the zero-shot scores of vision–language models, and text-prompted erasure of the artifact does not remove in-ROI shortcuts, whereas image-side synthetic insertion does.
3. **A closed-form account** (Sec. 6) of when in-ROI masking harms, why erasure fails for pathology-like artifacts, and why group reweighting is location-invariant; fitted per model to clean and correlated performance, it matches the held-out reversed-test results and location crossovers of real encoders (pre-registered).
4. **A remedy for frozen foundation-model features (U-MtE)** that needs no artifact annotation (only the ROI mask), a disease-protected variant, an erase-while-fine-tuning variant for fully fine-tuned CNNs and ViTs, with the evidence that erasure, not augmentation, is what works, and a delineation of the regimes in which it does not.
5. **A practical decision procedure**: a validation-driven selector and a decision rule based on where the artifact sits and what carries the shortcut, benchmarked on the real-artifact cohorts against DFR, JTT, LEACE, SPLINCE, LaMa inpainting and text-prompted erasure (GroupDRO only in the controlled dermoscopy sweeps), with gaming-aware reporting.
6. **Reproducibility**: pre-registration before every experiment, Holm-corrected primary claims, replication of the original pilot (73/88 numbers matched), public code and a two-command tool.

2. Related work

Artifacts in dermoscopy. Bissoto et al. [3] annotated seven artifact types in ISIC images and showed that debiasing is harder than it looks; Nauta et al. [18] neutralised coloured calibration patches by inpainting and retraining—an artifact that a colour rule localises and that lies largely outside the lesion. Hair occlusion degrades classifiers trained on HAM10000 [27], and removing dermoscopic dark-corner artifacts is not clearly beneficial, whereas adding synthetic ones to the training set improves classification [21]; both treat an artifact’s presence, not its position relative to the lesion. MaskMedPaint [13] inpaints regions *outside* the classification-relevant area. None of these treats artifact location relative to the ROI as a variable.

Ultrasound. In fetal ultrasound, calipers and on-screen text act as shortcuts even for segmentation [16]; we study them for classification in thyroid and ovarian ultrasound, where they lie on the nodule or tumour.

Chest radiography. Hospital-specific signals limit the generalisation of pneumonia classifiers [32]. Drain-driven pneumothorax shortcuts are documented on NIH-CXR14 and CheXpert [19, 12]; recent work localises them to late layers of probed encoders [20] and mitigates them by prevalence-equalised recalibration [14], which requires the artifact label at test time.

Group robustness and erasure. Reweighting and GroupDRO [23], last-layer retraining [15] and linear concept erasure [22, 1] require artifact labels or paired data; erasure from unpaired labels removes label-correlated signal when the concept is correlated with the label, which is why we build pairs by insertion. Task-preserving linear erasure (SPLINCE; 11) and projection of biased directions in vision–language models [5] are the closest erasure methods; both require concept labels or prompts, whereas our concept subspace is estimated from synthetic insertions. Random erasing and occlusion augmentation [34, 8] use similar overlays as training noise; we show that this does not remove the shortcut. Test-time selection [2] steers dermoscopy models with user keypoints.

Foundation models. Classifiers trained on frozen foundation-model features inherit shortcut-like behaviour that pooled multi-institutional evaluation can hide [10], and shortcut testing [4] probes whether a model’s performance gaps stem from encoding a sensitive attribute. We ask where in the image the shortcut sits and remove it from the frozen representation.

3. Data

Controlled dermoscopy cohort. ISIC 2018 Task 1/2 images with manual lesion masks and melanoma labels from the union of the ISIC 2016 and 2017 tables (three images with conflicting labels dropped); perceptual-hash (Hamming ≤ 8) and lesion grouping; 2,437 images for which a 65×19 px ruler (0.46% of a 518×518

image) can be placed at 0/25/50/75/100% overlap with the lesion within 10 percentage points (train/val/test 1,849/371/371 before common support).

Real hair cohort (ISIC 2019). All 25,331 ISIC 2019 training images; hair/ruler masks from the archive of Wegley et al. [28]; lesion masks manual for HAM10000 [25] and predicted elsewhere by a U-Net trained on ISIC 2018 Task 1. Two traps share one hair-free group (≤ 30 hair pixels at native resolution): Trap A contains hair with overlap $r = |\text{hair} \cap \text{lesion}|/|\text{hair}| \geq 0.5$, Trap B $r < 0.1$. Cells are subsampled to the hair-free source mix (HAM/BCN/MSK) within each label, giving Trap A 704/157/2,738/989 and Trap B 704/157/4,327/930 images (hair-free benign/melanoma, hair benign/melanoma); five seeds of five-fold group-safe cross-validation (leakage groups: lesion identifiers and pHash); 172 reversed-test melanomas per trap and seed.

Thyroid ultrasound. TN3K images with TNCD benign/malignant labels (3,493 images; 1,210 malignant) and manual nodule masks. Sonographer calipers were segmented by a frozen rule-based detector (dotted measurement lines and ‘+’ ends; audit in Sec. 11). Trap A: calipers with $r \geq 0.5$ (1,150 images); Trap B: $r < 0.1$ (326); caliper-free: 1,801. No patient identifiers are published; leakage groups are pHash chains (stricter embedding-based groups in Sec. 9.6).

Ovarian tumour ultrasound. MMOTU OTU_2d [33] (CC BY 4.0): 1,202 images with tumour masks, labelled benign cystic (chocolate cyst, serous cystadenoma, simple cyst) versus lesions with solid components (teratoma, theca-cell tumour, mucinous cystadenoma, high-grade serous carcinoma). The thyroid detector is applied unchanged: Trap A 341/276, Trap B 91/103, caliper-free 170/178 (benign-cystic/solid).

Capsule endoscopy. SEE-AI frames [31] (CC BY 4.0) containing only erosions (3,550) or only polyp-like lesions (1,931); the ROI is the union of the expert lesion boxes. Contamination (debris, bubbles, bile) is segmented by a linear probe on DINOv2 patch tokens trained on 2,160 expert-masked frames (held-out pixel accuracy 0.92, IoU 0.62; expert masks are used where available). Debris-free: $< 3\%$ of the frame; Trap A: debris $\geq 10\%$ with $r \geq 0.5$; Trap B: $r < 0.1$. Leakage groups: blocks of 100 consecutive frame numbers plus pHash duplicates.

Chest radiography. NIH ChestX-ray14 [26]; lung ROIs from the TorchXRayVision PSPNet [6]. *Controlled cohort:* 1,906 pneumothorax images without a chest drain (NEATX annotations) and 5,718 *No Finding* images from other patients; a 100×12 px radio-opaque tube is placed at controlled overlap with the lungs (7,610 images in common support). *Real drains* (in-ROI only): 29,687 images; drains are labelled by NEATX for pneumothorax-positive images and by a drain detector (precision 0.95) for negatives; environments are built within view position (AP/PA); five patient-grouped folds.

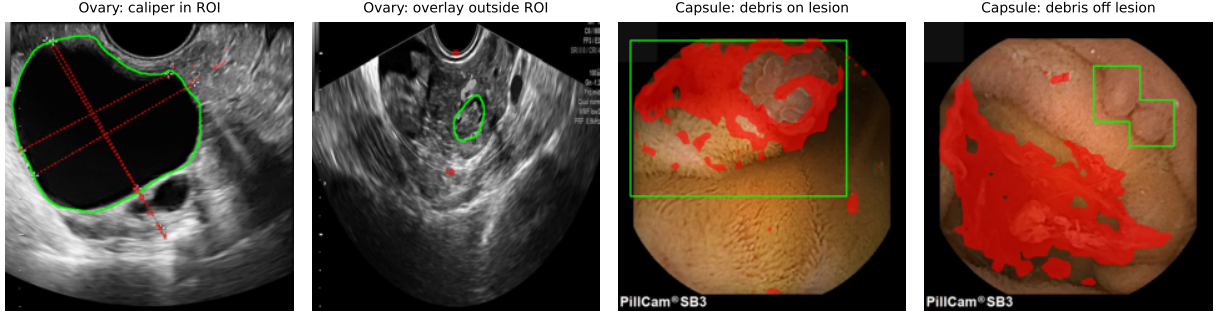


Figure 1: Real in-ROI and out-of-ROI artifacts (openly licensed cohorts). Green: ROI; red: artifact mask. Left: MMOTU ovarian tumours (caliper inside the tumour; overlay outside). Right: SEE-AI capsule frames (debris on and off the lesion box).

4. Problem setup and protocol

Let $Y \in \{0, 1\}$ be the diagnosis and $A \in \{0, 1\}$ the presence of an artifact. A *trap* trains with $P(A=1 | Y=1) = 0.9$ and $P(A=1 | Y=0) = 0.1$ and tests on a correlated environment with the same rates, a *reversed* environment with the rates swapped (0.1/0.9), and a clean environment in which the artifact is independent of the label (real traps: present in half of each class; controlled sweeps: absent). A model that uses A gains on the correlated test and collapses on the reversed one; the primary metric is reversed-test AUROC, and clean AUROC measures the cost of a mitigation. The overlap $r = |\text{artifact} \cap \text{ROI}|/|\text{artifact}|$ is the controlled variable. All backbones are frozen and only linear heads are fitted (an end-to-end fine-tuning check is reported in Sec. 9.2); regularisation strength and the decision threshold (balanced accuracy) are selected on clean validation data only. Uncertainty is a hierarchical paired bootstrap that resamples training seeds (or cross-validation folds), then paired test images within each, computes the AUROC difference inside each cluster and averages; predictions of different models are never pooled into one ROC curve. Splits are grouped by lesion/patient identifier where published and otherwise by perceptual hash (and frame blocks); a stricter embedding-based grouping is a sensitivity analysis (Sec. 9.6).

5. Mask-then-erase with a generic insertion library

Insertion pairs. Removing an in-ROI artifact requires its pixels; inserting one does not. Given a frozen encoder f and an overlay generator $\mathcal{G}$, every training image x_i yields a counterfactual $x_i^+ = \mathcal{G}(x_i; \xi_i)$ with random parameters ξ_i and random position, half of the elements centred inside the ROI. Because insertion ignores the label, the differences $\Delta_i = f(x_i^+) - f(x_i)$ are independent of Y .

Generic library. $\mathcal{G}$ is one procedural family used unchanged for every modality and artifact: thin curvilinear strokes, thick bands, solid or translucent patches, blobs, dashed lines and small glyphs, with random colour

(half greyscale), opacity and width. It contains no ruler, hair, caliper or debris template; it is a prior over “foreign overlays”, not a model of any particular artifact.

Erasure. The insertion eraser removes the span of the differences:

$$\Sigma_\Delta = \frac{1}{n} \sum_i \Delta_i \Delta_i^\top, \quad U_k = \text{top-}k \text{ eigvecs}(\Sigma_\Delta), \quad P(z) = z - U_k U_k^\top (z - \mu), \quad (1)$$

with k chosen *without labels* as the smallest rank whose held-out paired-difference energy $\mathbb{E}\|P\Delta\|^2/\mathbb{E}\|\Delta\|^2$ falls below 0.10 (fixed before any experiment).

Mask-then-erase (U-MtE). Masking and erasure have complementary reach: masking removes any artifact outside the ROI but none inside; erasure acts on the representation regardless of location. U-MtE therefore computes the pairs *on masked images* ($\Delta_i = f(m(x_i^+)) - f(m(x_i))$) so that the erased subspace targets exactly what masking leaves behind, and fits the head on $P(f(m(x)))$. When image-level artifact labels exist, the head is group-balanced (U-MtE_{bal}).

Disease protection. If the artifact resembles the pathology (intestinal debris versus erosion fibrin), U_k can contain disease evidence. Let W be an orthonormal basis of logistic label directions fitted on artifact-free training images; we erase $U'_k = \text{orth}((I - WW^\top)U_k)$ instead, which guarantees $w^\top P(z) = w^\top z$ for every $w \in \text{span}(W)$ (cf. task-preserving erasure, 11).

Scope. U-MtE is a representation-level intervention: it acts on the features of a frozen encoder, the dominant way in which foundation models are deployed on small medical datasets, and needs only the encoder, the ROI mask and the generator. An end-to-end analogue is evaluated as a secondary check (Sec. 9.2).

Tool. The released package exposes insertion, erasure and a patch-token artifact probe for any frozen backbone (ViTs and CNNs) through a two-call API.

6. Why location matters: a linear-Gaussian account

Let a frozen encoder yield a disease block with class-separation signal-to-noise ratio S and an artifact block with SNR A per unit of artifact presence; training uses $P(A=1 \mid Y=1) = p_1$, $P(A=1 \mid Y=0) = p_0$ and the reversed test swaps them. For the LDA-optimal linear head the artifact’s effective training SNR is $\tilde{A} = (p_1 - p_0)^2 A / (1 + p_1(1 - p_1)A)$ and

$$\text{AUROC}_{\text{rev}}(S, \tilde{A}) = \Phi\left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}}\right), \quad (2)$$

increasing in S and decreasing in $\tilde{A}$ (verified by simulation to within 0.009 AUROC; Fig. 2). Masking maps $(S, \tilde{A}) \mapsto (S_m, \tilde{A}_m)$. Out of the ROI, $\tilde{A}_m = 0$; inside, $\tilde{A}_m \geq \tilde{A}$. Hence (i) in-ROI masking harms exactly

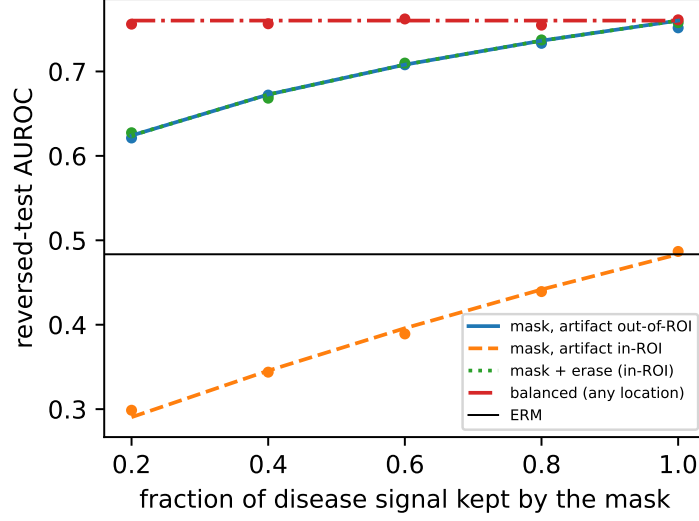


Figure 2: Reversed-test AUROC predicted by the linear-Gaussian model (lines) and simulated with logistic heads (points) as masking removes a growing share of disease signal ($S = 1$, $A = 2$, 90/10 training). Generated by `scripts/analysis/theory_sim.py`.

when the mask removes disease-informative context ($S_m < S$) and helps only by denoising ($S_m > S$); (ii) for any S_m the benefit of masking is larger out of the ROI than inside it—the crossover is always positive; (iii) erasing a subspace that contains the artifact direction sets $\tilde{A} \rightarrow 0$ but scales S by $1 - \rho^2$, where ρ is its overlap with the disease direction, so erasure beats in-ROI masking unless the artifact resembles the pathology, which disease protection prevents; (iv) group balancing removes the artifact–label association and everything correlated with it, giving $\Phi(\sqrt{S/2})$ irrespective of location; (v) a shortcut carried by correlates that neither the mask nor pixel-level erasure touch can only be removed by label-based reweighting. Each consequence is borne out in Sections 7–10.

Does the account match real encoders? For any test composition $q_y = P(A=1 | Y=y)$ the same head gives $\text{AUROC} = \Phi(\Delta/\sqrt{\sigma_1^2 + \sigma_0^2})$ with $\Delta = S + (p_1 - p_0)(q_1 - q_0)A/(1 + vA)$, $\sigma_y^2 = S + w_a^2(1 + Aq_y(1 - q_y))$, $w_a = \sqrt{A}(p_1 - p_0)/(1 + vA)$ and $v = [p_1(1 - p_1) + p_0(1 - p_0)]/2$. In a pre-registered test we fitted (S, A) for every trained model to its clean and correlated AUROC *only* and compared the implied reversed-test AUROC with the held-out observation; no parameter is shared across models or tuned (Fig. 3). Over 148 ERM and masking models (real traps, chest drains, controlled sweeps) the mean absolute error is 0.039 AUROC ($r = 0.92$), against 0.234 for “reversed = clean” and 0.108 for a linear-symmetry heuristic (2-clean – correlated). The implied location crossover agrees with the observed one across 14 cohort–backbone pairs ($r = 0.997$, mean absolute error 0.014, 13/14 signs; the miss is a chest-radiograph cell with an observed crossover of -0.002), the implied mask gains agree with the controlled sweeps ($r = 0.965$, 42/45 signs), and

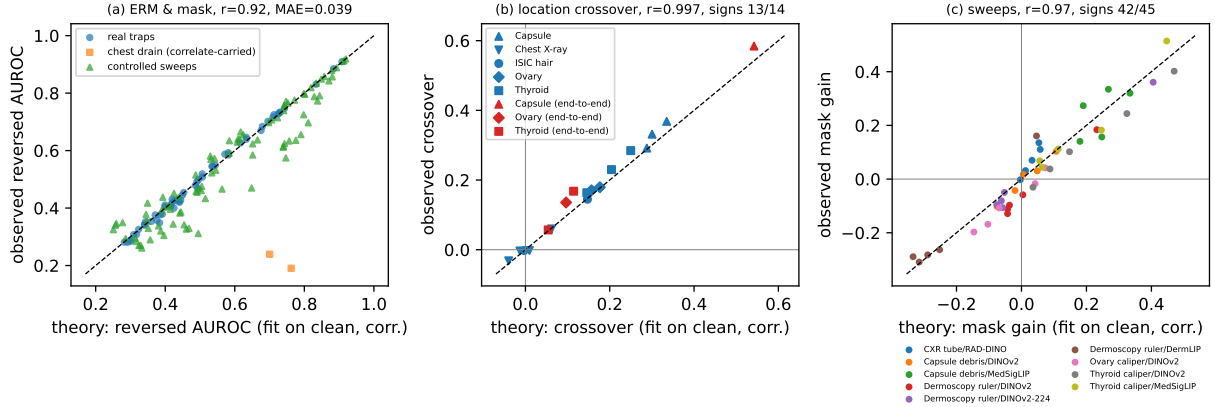


Figure 3: Pre-registered test of the linear-Gaussian account on real encoders. (S, A) are fitted to each model’s clean and correlated AUROC; the reversed-test results are held out. (a) Implied vs observed reversed-test AUROC (ERM and masking). (b) Location crossover per cohort and backbone; red: end-to-end fine-tuned networks, outside the model’s assumptions. (c) Mask – ERM gain along the controlled overlap sweeps. Dashed: identity. Generated by `scripts/analysis/theory_predict.py`.

the agreement extends to end-to-end fine-tuned networks (4/4 crossover signs), which violate the model’s assumptions. Because our pre-registered criterion required every crossover sign to agree, we still call the model a qualitative account. Its one clear failure is the chest-drain cohort (implied 0.70–0.76, observed 0.19–0.24): the clean and correlated environments do not expose a shortcut carried by correlates of the drain, regime (v).

7. Results: dermoscopy

7.1. Masking reverses with location (controlled)

With a synthetic ruler whose overlap with the lesion is swept from 0 to 100% (2,437 ISIC 2018 images, DINOv2 ViT-B/14 at 518 px, three seeds), ROI masking improves reversed-environment AUROC over ERM by $+0.184$ [$+0.146, +0.224$] when the ruler lies outside the lesion and *lowers* it by -0.097 [$-0.138, -0.056$], -0.112 [$-0.155, -0.068$] and -0.129 [$-0.170, -0.087$] at 50, 75 and 100% overlap (location interaction $+0.313$ [$+0.272, +0.353$]; Table 1). The sign flip is reproduced at 224 px ($+0.361$ at 0%; -0.095 at 100%) and under randomised ruler colour, opacity, ticks and scale. At 0% overlap the masked model’s correlated and reversed AUROC coincide (0.849/0.849): the shortcut is removed. At 100% the masked model’s correlated AUROC exceeds ERM’s (0.961 vs 0.937): it uses the ruler *more*.

Insert-then-erase in this table is the artifact-specific precursor of U-MtE: its insertion pairs use templates of the artifact type (rulers) and it erases from unmasked features. U-MtE (Sec. 5) replaces the templates by one generic overlay library for every modality and erases after masking; its dermoscopy results are in Sec. 10.

Table 1: Synthetic ruler, ISIC 2018, DINOv2 ViT-B/14 @518: AUROC and reversed-test Δ vs ERM [95% CI]

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	Δ rev@0	Δ rev@50	Δ rev@100
ERM	—	0.797	0.665	0.793	0.573	0.792	0.515			
ROI masking	ROI mask (train+test)	0.849	0.849	0.829	0.476	0.825	0.386	+0.184 [+0.146, +0.224]	-0.097 [-0.138, -0.056]	-0.129 [-0.170, -0.087]
Inpainting (oracle)	artifact pixels (train+test)	0.810	0.806	0.809	0.783	0.807	0.759	+0.141 [+0.120, +0.162]	+0.211 [+0.187, +0.234]	+0.243 [+0.218, +0.269]
Group-balanced	image-level A (train)	0.814	0.802	0.815	0.792	0.806	0.796	+0.137 [+0.109, +0.168]	+0.219 [+0.188, +0.252]	+0.281 [+0.248, +0.315]
DFR	image-level A (val)	0.786	0.789	0.787	0.794	0.789	0.794	+0.124 [+0.091, +0.160]	+0.221 [+0.184, +0.261]	+0.279 [+0.240, +0.320]
LEACE (paired, oracle renders)	paired renders (train)	0.829	0.832	0.810	0.810	0.809	0.809	+0.167 [+0.139, +0.197]	+0.237 [+0.212, +0.263]	+0.294 [+0.265, +0.325]
Repair consistency	artifact pixels (train)	0.828	0.707	0.822	0.616	0.814	0.558	+0.042 [+0.015, +0.069]	+0.043 [+0.014, +0.072]	+0.042 [+0.012, +0.073]
Insert-then-erase (artifact templates)	artifact templates	0.848	0.844	0.850	0.833	0.847	0.818	+0.179 [+0.152, +0.206]	+0.260 [+0.229, +0.291]	+0.303 [+0.271, +0.336]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.841	0.842	0.843	0.840	0.844	0.842	+0.177 [+0.151, +0.205]	+0.267 [+0.238, +0.298]	+0.327 [+0.293, +0.362]
Insert-then-erase rank-1 (ablation)	artifact templates	0.811	0.799	0.809	0.770	0.806	0.730	+0.134 [+0.115, +0.154]	+0.197 [+0.174, +0.220]	+0.215 [+0.190, +0.240]
Insertion augmentation	artifact templates	0.814	0.726	0.812	0.654	0.810	0.603	+0.061 [+0.049, +0.074]	+0.081 [+0.068, +0.094]	+0.088 [+0.077, +0.099]

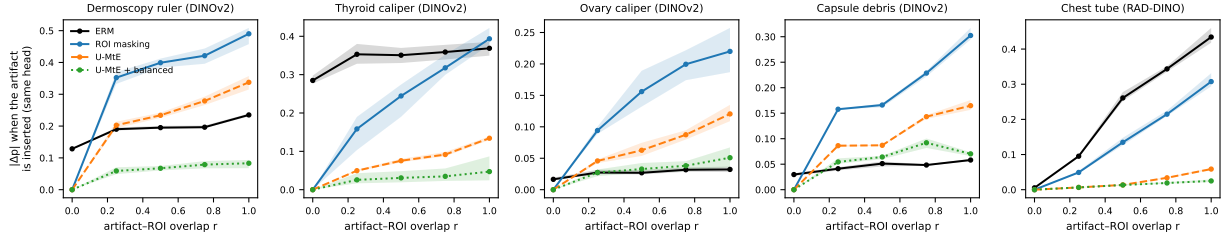


Figure 4: Mechanism: same-head counterfactual sensitivity $|p(x+\text{artifact}) - p(x)|$ versus the artifact’s overlap with the ROI in the five controlled sweeps (mean over seeds; band: range over seeds). Masking removes the out-of-ROI artifact ($r = 0$) but increasingly relies on an in-ROI one; erasure and balancing remove that reliance. Generated by `scripts/analysis/mechanism_figure.py`.

7.2. Mechanism: retention, not occlusion or context

Three explanations were tested separately. *Occlusion*: with the ruler present but uncorrelated with the label (50/50), masking helps at every overlap (+0.036, +0.040, +0.039 at 0/50/100%); covering lesion pixels costs nothing measurable, so the harm requires the correlation. *Lost context*: dilating the mask at 50% overlap *increases* harm ($-0.097 \rightarrow -0.138 \rightarrow -0.154$ for margins 0/10/25 px) as the margin restores ruler pixels (retention 0.50 $\rightarrow$ 0.73 $\rightarrow$ 0.95); at 100% overlap, where retention is already 1, harm is flat (-0.129 to -0.144) although the ruler’s share of visible pixels falls threefold. *Amplified reliance*: masking doubles the same-head counterfactual sensitivity to the ruler ($|\Delta p|$ 0.235 $\rightarrow$ 0.491 at 100%). Harm is largest for small lesions (-0.364 vs $+0.064$ for large lesions at 100%), where the ruler is a larger share of what survives the mask. Figure 4 shows the same retention mechanism in all five controlled sweeps: with the head held fixed, inserting the artifact changes the masked model’s prediction more the deeper the artifact lies in the ROI. At full overlap this sensitivity exceeds that of ERM for dermoscopic rulers (0.491 vs 0.235), ovarian calipers (0.218 vs 0.032) and capsule debris (0.302 vs 0.058) and matches it for thyroid calipers (0.393 vs 0.369). In chest radiography ERM ignores a tube outside the lungs ($|\Delta p| = 0.005$ at $r = 0$), which is why masking has nothing to remove there. U-MtE lowers the masked model’s sensitivity in every sweep (full overlap: 0.338, 0.134, 0.120, 0.164, 0.059), and U-MtE with group balancing lowers it most (≤ 0.083).

7.3. Real hair (ISIC 2019), pre-registered

Two traps share one hair-free group (at most 30 hair pixels): Trap A contains hair lying mostly inside the lesion ($r \geq 0.5$), Trap B hair mostly outside ($r < 0.1$); cells are matched to the hair-free source mix, and five seeds of five-fold group-safe cross-validation give 172 reversed-test melanomas per trap. Masking helps in Trap B (+0.107 [+0.078, +0.135]), harms in Trap A (−0.037 [−0.057, −0.019], negative in all five seeds) and the crossover is +0.145 [+0.105, +0.181]. The harm holds in BCN images (−0.051 [−0.075, −0.027]); in HAM it is not distinguishable from zero (−0.009 [−0.036, +0.017]). Oracle hair inpainting helps in both traps (+0.103, +0.096). Group-balanced training and DFR, which need only an image-level hair label, give the largest gains (+0.225/+0.251 in Trap A, +0.209/+0.215 in Trap B). Paired linear erasure recovers less than half of that (+0.086/+0.081). The magnitude of the in-ROI harm depends on the definition of the comparison group: with a looser hair-free group ($< 0.1\%$ of pixels) and only heavily haired images, masking gives no benefit but no measurable harm inside the lesion (+0.012 [−0.042, +0.062]) while still helping outside (+0.178); the benefit shrinks to zero as more of the hair lies inside the lesion.

With the dermatology foundation model DermLIP, the controlled ruler sweep reverses as with DINOv2 (+0.161 at no overlap, −0.263 to −0.309 at 25–100%), but on real hair masking harms at both locations (Trap A −0.085 [−0.107, −0.063], Trap B −0.088): with this encoder masking removes disease context regardless of where the hair lies, so no location gap appears.

7.4. What the real-hair shortcut is made of

Removing hair pixels at test time only closes 44–47% of the ERM clean–reversed gap: most of what a frozen encoder exploits in a real hair trap is not the hair pixels but what accompanies hair (hair-bearing images differ in sex, age and body site). This bounds every pixel-level method, including oracle inpainting and U-MtE, and explains why image-level-label methods, which remove everything correlated with the label, recover more.

7.5. Reversed-environment gains can be gamed

Erasing the artifact concept from unpaired image-level labels yields the largest reversed gain in the study (+0.326 in Trap A) while *inverting* the correlated/reversed ordering (0.618 vs 0.817). Post-hoc prevalence-equalised recalibration [14] behaves the same way under the trap’s 90/10 shift (+0.427; correlated 0.433 vs reversed 0.918). Both remove diagnosis-correlated signal or over-correct the group prior rather than removing reliance on the artifact; we therefore read every reversed gain together with clean AUROC and the correlated/reversed ordering.

8. Results: ultrasound, capsule endoscopy and chest radiography

8.1. Thyroid ultrasound: sonographer calipers

TN3K images with TNCD labels (3,493 images, 1,210 malignant) carry measurement calipers that a sonographer places on or around the nodule. Calipers were segmented by an audited rule-based detector (image-level precision and recall 0.86–0.89). With the E13 protocol unchanged, masking the nodule helps greatly when the calipers lie outside it (Trap B: +0.341 [+0.320, +0.363]) and much less when they lie inside it (Trap A: +0.111 [+0.097, +0.124]); the crossover is +0.230 [+0.204, +0.255] with DINOv2 and +0.163 [+0.132, +0.194] with MedSigLIP. After masking, the Trap A model still relies on the calipers (correlated 0.902 versus reversed 0.400). With synthetic calipers swept across the nodule boundary (1,663 caliper-free images), masking gains +0.402 [+0.347, +0.458] at no overlap and -0.030 [-0.079 , +0.016] at full overlap (interaction +0.432 [+0.377, +0.492]): the entire benefit of masking disappears, without significant harm.

8.2. Capsule endoscopy: intestinal debris

SEE-AI frames (erosion 3,550 versus polyp-like 1,931; expert lesion boxes as ROI) with contamination masks from a patch-token probe trained on 2,160 expert-masked frames (held-out pixel accuracy 0.92, IoU 0.62). Leakage groups are contiguous blocks of 100 frame numbers plus perceptual-hash duplicates (no video identifiers are published). Masking helps against out-of-ROI debris (+0.465 [+0.440, +0.490]) and much less against debris on the lesion (+0.097), a crossover of +0.368 [+0.340, +0.397]. With synthetic debris on the 796 cleanest frames the reversal appears: +0.103 [+0.060, +0.149] at no overlap, -0.102 [-0.162 , -0.040] at full overlap (interaction +0.205 [+0.150, +0.259]).

8.3. Chest radiography: a boundary case

With a synthetic tube swept into the lung fields of NIH pneumothorax radiographs (RAD-DINO), the encoder does not use an extra-pulmonary tube at all (reversed 0.913 versus clean 0.915 at no overlap), so masking has nothing to remove; inside the lungs the tube is used (reversed 0.614) and lung masking only partly removes it (0.725). On real devices (RANZCR-CLiP linked to NIH; central lines and endotracheal tubes) masking helped for both locations. The location-dependent *harm* of masking is therefore not universal; what is universal in our data is that masking leaves an in-ROI shortcut largely intact.

9. Results: natural distribution, fine-tuning and label robustness

9.1. Unaltered test distributions

Without any resampling, in-ROI artifacts are naturally associated with the label in every cohort (thyroid calipers: 37.6% of benign vs 24.1% of malignant nodules; in-lesion hair: 31% of melanomas vs 20% of benign lesions; capsule debris on the lesion: 23% of polyp-like lesions vs 10% of erosions). We trained on natural

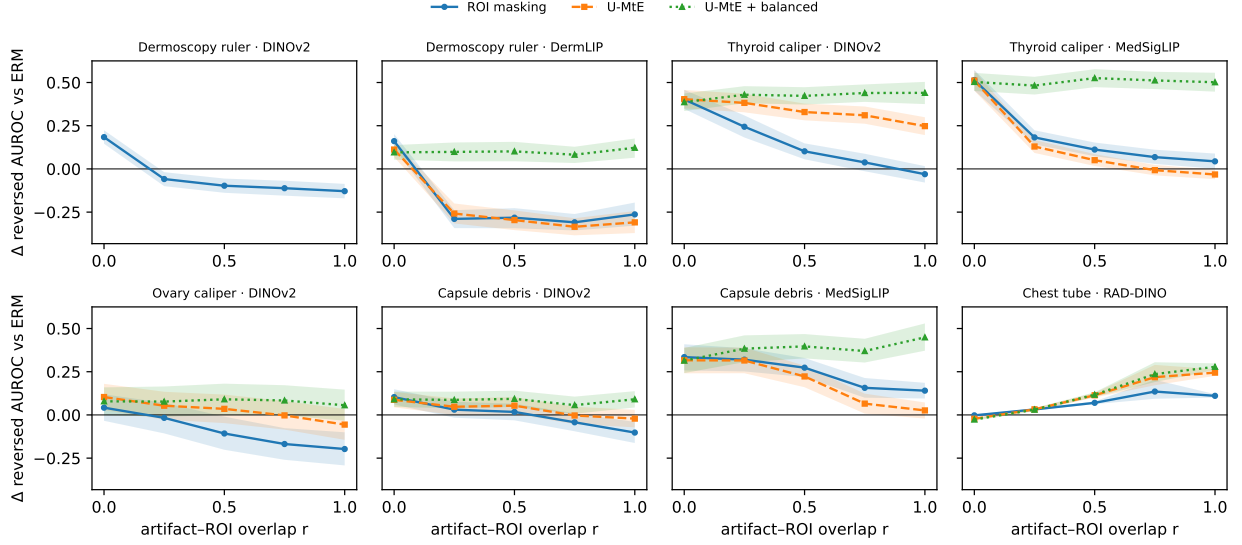


Figure 5: Controlled overlap sweeps: change in reversed-test AUROC versus ERM as the synthetic artifact moves into the ROI (95% CIs shaded). ROI masking (blue) declines with overlap in every modality except chest radiography, where the encoder ignores extra-pulmonary tubes; U-MtE with group balancing (green) is flat. Generated by `scripts/make_figures.py`.

training data and tested on untouched data: the official TN3K test split, each ISIC source held out in turn (a real change of hospital and device), and held-out capsule frame blocks. On the shortcut-conflicting pairs, masking *lowers* AUROC for thyroid (-0.102 [$-0.139, -0.066$]) and for two of the three held-out hospitals (BCN -0.016 [$-0.027, -0.005$], MSK -0.056 [$-0.084, -0.029$]); it helps for HAM ($+0.053$) and does not differ significantly from ERM on capsule. U-MtE with group balancing improves on masking for these pairs in every cohort ($+0.014$ to $+0.088$) and gives the best or tied-best cross-group AUROC in three of five settings (HAM 0.786, BCN 0.734 vs 0.735, capsule 0.947); plain group balancing is best on thyroid and MSK. Without artifact labels, U-MtE improves on masking for thyroid ($+0.051$) and HAM ($+0.023$) only.

Clinical secondary metrics. Table 2 reports AUPRC, Brier score, expected calibration error and sensitivity/specificity at the operating point fixed on validation data. On the official thyroid split masking lowers sensitivity from 0.734 to 0.347 while specificity rises from 0.619 to 0.888 (balanced accuracy $0.677 \rightarrow 0.618$); U-MtE recovers part of the sensitivity (0.425) and has the best AUPRC (0.638) and Brier score (0.203). For the held-out BCN hospital masking worsens both Brier score ($0.239 \rightarrow 0.286$) and calibration (ECE $0.279 \rightarrow 0.332$); for HAM, MSK and capsule it improves every metric, in line with the AUROC results.

9.2. End-to-end fine-tuning

Fine-tuning ResNet-50 end-to-end (five folds) reproduces the frozen-backbone findings. Capsule: masking harms in-ROI (-0.284 [$-0.347, -0.226$]) and helps out-of-ROI ($+0.301$), crossover $+0.584$ [$+0.469, +0.691$].

Table 2: Clinical secondary metrics on the unaltered test sets (Sec. 9): mean (SD) over training seeds. Sensitivity and specificity at the operating point fixed on clean validation data; ECE with 10 equal-width bins; AUPRC baseline = prevalence.

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sens	Spec
Thyroid (official split) ($\pi = 0.38$)	ERM	0.736 (0.009)	0.606 (0.012)	0.207 (0.005)	0.082 (0.008)	0.734 (0.047)	0.619 (0.061)
	mask	0.731 (0.006)	0.632 (0.007)	0.214 (0.006)	0.120 (0.026)	0.347 (0.095)	0.888 (0.041)
	U-MtE	0.742 (0.003)	0.638 (0.007)	0.203 (0.001)	0.085 (0.007)	0.425 (0.088)	0.854 (0.050)
	balanced	0.716 (0.007)	0.584 (0.012)	0.219 (0.004)	0.110 (0.007)	0.714 (0.052)	0.599 (0.061)
	U-MtE _{bal}	0.731 (0.004)	0.630 (0.007)	0.207 (0.001)	0.084 (0.006)	0.399 (0.069)	0.866 (0.034)
ISIC, BCN held out ($\pi = 0.23$)	ERM	0.786 (0.003)	0.586 (0.003)	0.239 (0.006)	0.279 (0.011)	0.842 (0.038)	0.519 (0.075)
	mask	0.769 (0.004)	0.571 (0.009)	0.286 (0.007)	0.332 (0.009)	0.860 (0.013)	0.439 (0.023)
	U-MtE	0.773 (0.004)	0.574 (0.008)	0.280 (0.009)	0.328 (0.012)	0.858 (0.039)	0.448 (0.089)
	balanced	0.781 (0.005)	0.585 (0.009)	0.228 (0.006)	0.257 (0.011)	0.811 (0.046)	0.554 (0.087)
	U-MtE _{bal}	0.774 (0.004)	0.568 (0.008)	0.279 (0.011)	0.325 (0.016)	0.860 (0.027)	0.447 (0.051)
ISIC, MSK held out ($\pi = 0.19$)	ERM	0.770 (0.008)	0.477 (0.012)	0.172 (0.007)	0.161 (0.019)	0.608 (0.049)	0.786 (0.039)
	mask	0.790 (0.004)	0.512 (0.009)	0.146 (0.003)	0.115 (0.009)	0.593 (0.052)	0.824 (0.037)
	U-MtE	0.783 (0.004)	0.500 (0.008)	0.150 (0.003)	0.126 (0.009)	0.566 (0.073)	0.831 (0.049)
	balanced	0.766 (0.008)	0.462 (0.014)	0.178 (0.008)	0.170 (0.020)	0.626 (0.047)	0.755 (0.046)
	U-MtE _{bal}	0.774 (0.006)	0.482 (0.009)	0.154 (0.004)	0.131 (0.010)	0.594 (0.061)	0.798 (0.051)
ISIC, HAM held out ($\pi = 0.11$)	ERM	0.750 (0.005)	0.353 (0.005)	0.162 (0.006)	0.234 (0.011)	0.535 (0.092)	0.813 (0.068)
	mask	0.789 (0.002)	0.404 (0.006)	0.143 (0.005)	0.217 (0.011)	0.559 (0.112)	0.842 (0.063)
	U-MtE	0.799 (0.008)	0.406 (0.009)	0.145 (0.007)	0.226 (0.016)	0.573 (0.067)	0.846 (0.041)
	balanced	0.747 (0.017)	0.346 (0.019)	0.156 (0.014)	0.228 (0.015)	0.539 (0.078)	0.803 (0.076)
	U-MtE _{bal}	0.802 (0.005)	0.401 (0.009)	0.139 (0.006)	0.215 (0.021)	0.606 (0.054)	0.825 (0.037)
Capsule (held-out frames) ($\pi = 0.56$)	ERM	0.889 (0.019)	0.897 (0.046)	0.130 (0.016)	0.042 (0.011)	0.770 (0.062)	0.844 (0.055)
	mask	0.967 (0.011)	0.971 (0.014)	0.069 (0.013)	0.029 (0.011)	0.859 (0.032)	0.933 (0.025)
	U-MtE	0.962 (0.009)	0.967 (0.013)	0.081 (0.017)	0.065 (0.045)	0.871 (0.042)	0.897 (0.043)
	balanced	0.881 (0.024)	0.889 (0.053)	0.136 (0.017)	0.051 (0.009)	0.758 (0.087)	0.838 (0.059)
	U-MtE _{bal}	0.959 (0.008)	0.964 (0.012)	0.079 (0.007)	0.044 (0.012)	0.866 (0.048)	0.897 (0.052)

Thyroid: the crossover points the same way (+0.057 [−0.021, +0.130], five folds), and an end-to-end analogue of U-MtE—an invariance penalty between masked images with and without a generic overlay—improves on masking by +0.143 [+0.099, +0.188] without shortcut flipping (reversed 0.679, correlated 0.703), matching the best label-using arm. It fails on capsule (−0.021), where the artifact resembles the pathology. Ovary (1,202 images) is underpowered: U-MtE end-to-end has the best reversed, correlated and clean AUROC, with CIs overlapping masking. A fine-tuned ViT-S reproduces the crossover on thyroid (+0.168 [+0.074, +0.265]), but the end-to-end U-MtE analogue does not improve on masking with this architecture (+0.009 [−0.021, +0.040]). Applying U-MtE post hoc to the penultimate features of the masked, fine-tuned ResNet-50 does not help either (pre-registered; +0.004 [+0.000, +0.012] over the same features without erasure): after task training the shortcut is encoded along task-specific directions that generic overlays do not move. For fully fine-tuned networks the subspace must be estimated and removed *during* training. *Erase-while-fine-tuning* (pre-registered; no artifact annotation) projects a running

estimate of the overlay-shift subspace of the network’s own penultimate features out before the head at every step and at test time, and adds feature invariance and prediction consistency between masked images with and without a generic overlay. On thyroid it improves on masking by $+0.157$ [$+0.114, +0.205$] with ResNet-50 and $+0.143$ [$+0.103, +0.187$] with ViT-S (reversed AUROC 0.693 and 0.608 vs 0.536 and 0.465), whereas each component alone works for only one architecture (projection + invariance: ResNet-50 $+0.160$ [$+0.114, +0.207$], ViT-S $+0.061$ [$-0.005, +0.141$]; prediction consistency: ResNet-50 $+0.007$ [$-0.014, +0.029$], ViT-S $+0.062$ [$+0.031, +0.087$]). With ResNet-50 the correlated AUROC falls slightly below the reversed one (0.661 vs 0.693), a mild over-correction. On ovary (ResNet-50) it is positive but not significant ($+0.094$ [$-0.011, +0.210$]; every fine-tuning contrast on this 1,202-image cohort is underpowered), and on capsule, where debris resembles erosion fibrin, it fails (-0.022 [$-0.046, +0.001$]), as does every annotation-free method.

9.3. Zero-shot vision–language models

Without any task training, the foundation models’ own zero-shot scores (similarity to “benign” versus “malignant” text prompts) are already swayed by artifacts: on capsule frames the correlated and reversed AUROC differ by 0.40. Masking changes these zero-shot scores exactly as the location law predicts: for MedSigLIP it leaves reversed AUROC unchanged for in-ROI artifacts but improves it for out-of-ROI artifacts (thyroid $+0.013$ vs $+0.116$, crossover $+0.103$ [$+0.079, +0.127$]; capsule crossover $+0.049$ [$+0.024, +0.076$]), and it harms more for in-ROI artifacts on ovarian ultrasound (crossover $+0.181$ [$+0.135, +0.228$]) and for DermLIP on hair ($+0.051$ [$+0.031, +0.071$]). The location effect is therefore a property of the representation, not of the classifier trained on top of it.

9.4. Robustness to artifact-label definitions

Artifact masks for calipers come from a detector (visual audit: 95% of thyroid detections lie on real calipers; 92–97% of caliper-free images contain no visible caliper). Re-running the thyroid and ovary analyses with large-marker-only (≥ 50 px) and strict-location ($r \geq 0.7$ / $r < 0.05$) definitions leaves the crossover (thyroid $+0.215$ to $+0.262$; ovary $+0.170$ to $+0.195$) and the U-MtE_{protect} gain over masking (thyroid $+0.217$ to $+0.282$; ovary $+0.043$ to $+0.067$) significant in every variant; both grow under stricter definitions, as expected if detector errors dilute rather than create the effects.

9.5. Backbone scale

Replacing DINOv2 ViT-B/14 (86 M parameters) by ViT-S/14 (21 M) or ViT-L/14 (304 M), with everything else fixed (pre-registered), the location crossover is positive in all nine cohort–size combinations (ViT-S / ViT-B / ViT-L: thyroid $+0.244$ / $+0.230$ / $+0.271$; ovary $+0.298$ / $+0.170$ / $+0.202$; capsule $+0.370$ / $+0.368$ / $+0.408$; every CI excludes zero), and protected U-MtE beats masking on thyroid and ovary at

Table 3: In-ROI artifacts (Trap A): reversed-test AUROC, with min(reversed, correlated) in parentheses (penalises shortcut flipping). Best robust value per column in bold. Generated by `scripts/make_main_table.py`.

Method	Needs	Hair/DINOv2	Hair/DermlIP	Thyroid/DINOv2	Thyroid/MedSigLIP	Thyroid/ConvNeXt	Ovary/DINOv2	Ovary/MedSigLIP	Capsule/DINOv2	Capsule/MedSigLIP	Capsule/ConvNeXt	Drain/RAD-DINO	Drain/DINOv2
ERM	—	0.49 (0.49)	0.67 (0.67)	0.29 (0.29)	0.28 (0.28)	0.35 (0.35)	0.41 (0.41)	0.42 (0.42)	0.59 (0.59)	0.59 (0.59)	0.55 (0.55)	0.19 (0.19)	0.28 (0.28)
ROI masking	ROI	0.45 (0.45)	0.58 (0.58)	0.40 (0.40)	0.35 (0.35)	0.36 (0.36)	0.55 (0.55)	0.43 (0.43)	0.68 (0.68)	0.70 (0.70)	0.64 (0.64)	0.24 (0.24)	0.29 (0.29)
Mask + overlay augmentation	ROI	0.46 (0.46)	—	0.42 (0.42)	0.38 (0.38)	0.37 (0.37)	0.53 (0.53)	—	0.70 (0.70)	0.72 (0.72)	0.65 (0.65)	0.24 (0.24)	0.29 (0.29)
U-MtE (ours)	ROI	0.51 (0.51)	0.51 (0.51)	0.62 (0.62)	0.55 (0.55)	0.49 (0.49)	0.56 (0.56)	0.64 (0.64)	0.57 (0.57)	0.47 (0.47)	0.56 (0.56)	0.27 (0.27)	0.28 (0.28)
U-MtE, protected (ours)	ROI + Y of A=0	0.49 (0.49)	0.59 (0.59)	0.62 (0.62)	0.61 (0.61)	0.55 (0.55)	0.59 (0.59)	0.57 (0.57)	0.67 (0.67)	0.67 (0.67)	0.64 (0.64)	0.27 (0.27)	0.28 (0.28)
Group-balanced	A	0.72 (0.72)	0.81 (0.81)	0.68 (0.68)	0.70 (0.70)	0.66 (0.66)	0.71 (0.71)	0.73 (0.73)	0.78 (0.78)	0.81 (0.81)	0.72 (0.72)	0.54 (0.54)	0.40 (0.40)
DFR	A	0.74 (0.74)	0.78 (0.78)	0.74 (0.66)	0.76 (0.72)	0.71 (0.67)	0.72 (0.69)	—	0.80 (0.80)	0.80 (0.80)	0.77 (0.77)	0.68 (0.68)	0.55 (0.55)
Mask + DFR	ROI + A	0.69 (0.69)	—	0.83 (0.74)	0.81 (0.76)	0.77 (0.72)	0.75 (0.75)	—	0.88 (0.88)	0.90 (0.90)	0.86 (0.86)	0.65 (0.65)	0.57 (0.57)
U-MtE + balanced (ours)	ROI + A	0.70 (0.70)	0.65 (0.65)	0.79 (0.79)	0.79 (0.79)	0.74 (0.74)	0.74 (0.74)	0.79 (0.79)	0.86 (0.86)	0.85 (0.85)	0.80 (0.80)	0.46 (0.46)	0.44 (0.44)

every size (ViT-S +0.181 [+0.146, +0.218] and +0.103 [+0.078, +0.128]; ViT-L +0.222 [+0.195, +0.256] and +0.060 [+0.043, +0.077]). Larger models exploit the in-ROI shortcut *more*: the ERM correlated–reversed gap grows from 0.547 to 0.621 (thyroid) and from 0.319 to 0.570 (ovary) between ViT-S and ViT-L, so scale does not remove it. On capsule, disease protection is less complete with ViT-L (−0.040 [−0.046, −0.035] versus masking).

9.6. Stricter leakage grouping

TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers, so the main analyses group near-duplicates by perceptual hash (and, for capsule, blocks of 100 consecutive frames). As a pre-registered sensitivity analysis we united these groups with connected components of mean-centred DINOv2 embeddings at the most aggressive cosine threshold that keeps every group below 5% of the cohort (thyroid: 3,455 → 3,081 groups, $\tau = 0.80$; ovary: 1,059 → 969, $\tau = 0.80$; capsule: 137 → 135, $\tau = 0.93$) and re-ran all primary contrasts. Eleven of twelve keep their sign and significance: the location crossover is +0.235 [+0.211, +0.259] (thyroid), +0.157 [+0.080, +0.235] (ovary) and +0.394 [+0.359, +0.430] (capsule), and protected U-MtE still beats masking on thyroid (+0.251 [+0.219, +0.279]) and ovary (+0.047 [+0.020, +0.072]). The exception is the small ovary advantage of erasure over augmentation, which loses significance (+0.020 [−0.011, +0.053]). Embedding grouping removes visually near-identical images of one patient or session; it cannot exclude differently looking images of the same patient.

10. Results: artifact-agnostic mask-then-erase

10.1. U-MtE repairs what masking leaves behind

Without any artifact example, artifact mask or artifact label—it uses only the ROI mask that masking already requires—U-MtE improves reversed AUROC over masking in the in-ROI regime of every modality: thyroid calipers +0.222 [+0.197, +0.251] (DINOv2) and +0.204 [+0.174, +0.237] (MedSigLIP); dermoscopic hair +0.052 [+0.044, +0.062]; synthetic capsule debris +0.081 [+0.050, +0.112]; synthetic calipers +0.278

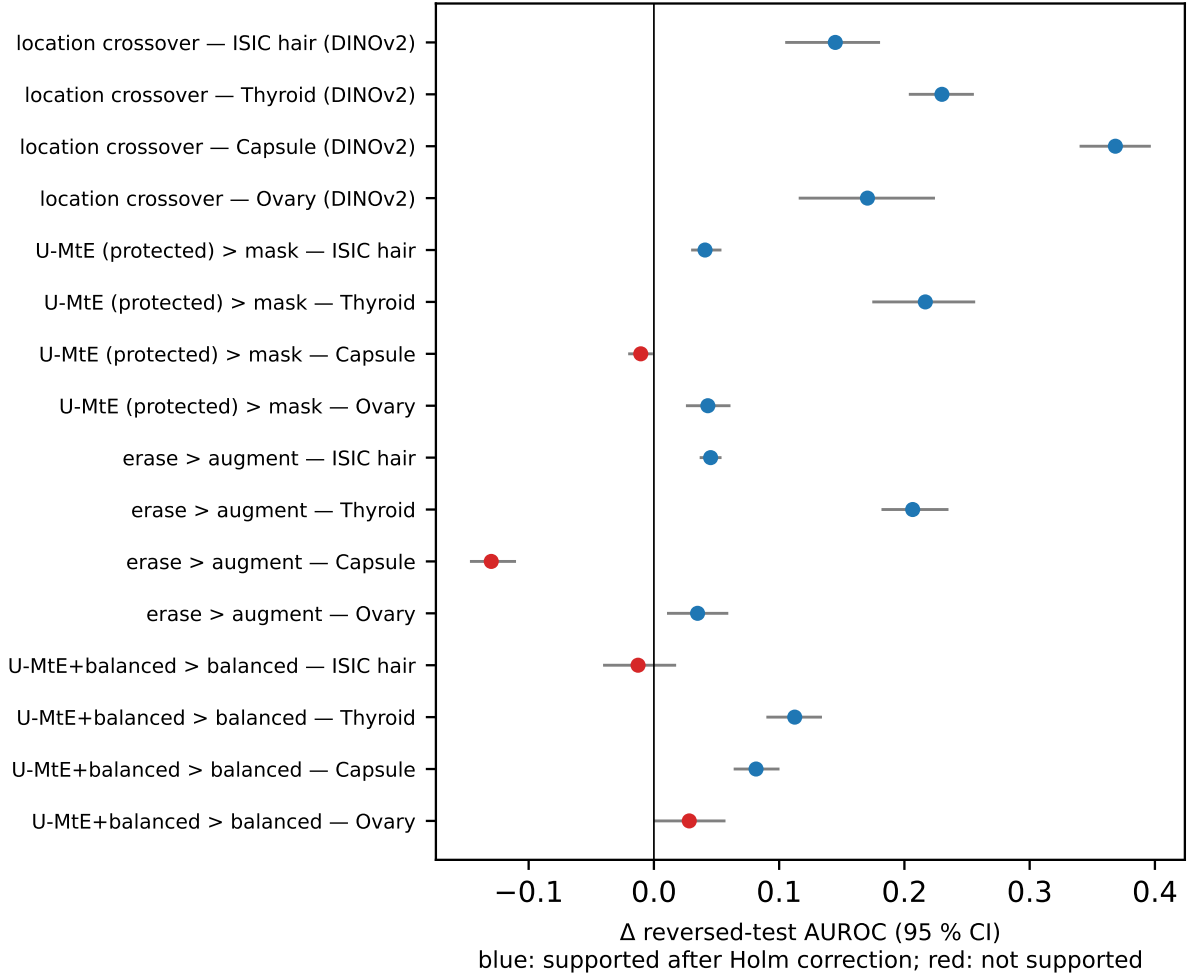


Figure 6: The 16 pre-registered primary tests with Holm correction (blue: supported; red: not supported). Generated by `scripts/analysis/primary_claims.py` and `scripts/make_figures.py`.

[+0.232, +0.325]; synthetic chest tubes +0.134 [+0.120, +0.151] (DINOv2 / RAD-DINO). With MedSigLIP the unbalanced method *loses* to masking in both controlled sweeps (debris -0.114 [$-0.147, -0.082$], calipers -0.076 [$-0.110, -0.043$]) although it wins on real thyroid calipers (+0.204); the balanced variant beats masking in all four controlled sweeps (+0.193 to +0.471) and has no location interaction in any of them. Disease protection turns the MedSigLIP debris failure into a gain (+0.061 [+0.032, +0.092] over masking) but not the caliper one (-0.041 [$-0.089, -0.005$]).

Sensitivity to the erasure rank. In a pre-registered ablation (Trap A, DINOv2), the thyroid gain over masking is +0.18 to +0.22 for every setting that erases at least 16 directions (held-out energy 0.5–0.99, or fixed $k \in \{16, 64\}$), but only +0.06 to +0.07 with one to four directions: the shift caused by overlays is multi-

dimensional. The pre-specified 90% rule reached the rank cap ($k = 64$) in all three cohorts. The protected variant never harms capsule by more than 0.010 and helps ovary at every rank (+0.007 to +0.043), whereas unprotected erasure harms capsule more the more directions it removes, as the overlap term ρ of Sec. 6 predicts.

10.2. Erasure, not augmentation

Using the same generic overlays as training augmentation (overlaid copies of a random half of the training images, in the masked view) gains at most +0.03 over masking; erasing their subspace instead gains +0.207 [+0.182, +0.235] (thyroid, DINOv2), +0.177 [+0.148, +0.211] (thyroid, MedSigLIP) and +0.045 [+0.037, +0.054] (hair) more than augmentation. Under the stricter near-duplicate grouping (Sec. 9.6) the thyroid advantage is unchanged (+0.192) but the small ovary advantage (+0.035) loses significance, so we regard erase > augment as established for thyroid and hair only.

10.3. When the artifact resembles the disease

On real capsule debris U-MtE is worse than masking (−0.115): erosions carry yellow-white fibrin that resembles debris, and a subspace estimated from real-debris templates is worse still (−0.395). Disease protection removes the failure (−0.010 [−0.020, −0.001] generic, −0.012 [−0.028, +0.003] template) at a cost of at most 0.012 where erasure already works (thyroid −0.005 [−0.036, +0.024], hair −0.012 [−0.020, −0.003]).

10.4. With image-level artifact labels

Adding group balancing, U-MtE_{bal} is the most robust model in thyroid ultrasound by the minimum of correlated and reversed AUROC (0.788 DINOv2, 0.792 MedSigLIP; DFR 0.659 and 0.719); in real capsule endoscopy masking plus last-layer retraining is higher (0.885 versus 0.857); on dermoscopic hair DFR alone is best (0.741) because most of the hair shortcut is not in the hair pixels (removing every hair patch with the oracle mask gains only +0.015). In both controlled sweeps the benefit of U-MtE_{bal} no longer depends on location (interaction −0.000 [−0.023, +0.023] capsule; −0.054 [−0.098, −0.004] thyroid), whereas that of masking does (+0.205, +0.432). The capsule findings replicate with MedSigLIP (crossover +0.332 [+0.296, +0.365]; masking plus DFR 0.901). All thyroid and capsule findings replicate with a CNN (ConvNeXt-Base): crossovers +0.284 and +0.292; U-MtE − mask +0.128 [+0.112, +0.145] and erase − augment +0.120 [+0.102, +0.139] on thyroid.

10.5. Label-free and erasure baselines

Among methods that need no artifact labels, U-MtE beats JTT [17] by +0.198 [+0.167, +0.231] (thyroid), +0.137 [+0.098, +0.179] (ovary) and +0.094 [+0.077, +0.109] (chest drains, RAD-DINO; not with DINOv2, −0.010); JTT on the unmasked view is best on hair. Task-preserving concept removal (SPLINCE; 11),

fitted with artifact labels in the whitened, minimal-distortion form of the original method, removes the artifact covariance at a large cost in disease signal (clean AUROC 0.56–0.78 versus 0.69–0.87 for masking) and still flips the shortcut on capsule endoscopy (reversed 0.868, correlated 0.683): when artifact and label are strongly associated in training, removing one removes much of the other. By $\min(\text{reversed}, \text{correlated})$ U-MtE_{bal} exceeds SPLINCE on thyroid, capsule and hair (0.788 vs 0.596; 0.857 vs 0.683; 0.704 vs 0.564); on chest drains SPLINCE is higher (0.609 vs 0.460) but below DFR (0.683).

10.6. Text-prompted erasure

With vision–language backbones an artifact direction can be obtained from text alone (“a thyroid ultrasound image with measurement calipers” minus “a thyroid ultrasound image”; biased-prompt construction, 5). Erasing that subspace after masking is a near no-op in every cohort (+0.001 to +0.008 over masking; MedSigLIP thyroid, ovary, capsule and DermLIP hair), whereas U-MtE with the same MedSigLIP features gains +0.204 [+0.174, +0.237] (thyroid) and +0.215 [+0.182, +0.246] (ovary). The direction along which a described artifact moves the *text* embedding is not the direction along which the real artifact moves the *image* embedding; synthetic insertions measure the latter directly.

10.7. A single procedure: validation-driven selection

Selecting, per fold, the candidate with the best group-aware validation score (image-level artifact labels needed on validation images only) beats masking in all five cohorts (+0.17 to +0.40), is within 0.02 of the best candidate in each, and beats DFR in three (thyroid +0.057, capsule +0.047, ovary +0.010); it chooses U-MtE_{bal} where the artifact is an overlay and plain balancing on hair and drains, where DFR remains better. A split-validation variant that admits DFR as a candidate is reported in the supplement.

10.8. Modern inpainting

Inpainting the detected caliper pixels with LaMa [24] and then masking is a strong baseline when artifact masks are available (thyroid +0.226 [+0.213, +0.238], ovary +0.123 [+0.089, +0.155] over masking). U-MtE_{protect}, which needs no artifact mask, matches it on thyroid (−0.009 [−0.045, +0.023]) and is below it on ovary (−0.080) and on dermoscopic hair, where published hair masks are available (−0.086); with image-level artifact labels, U-MtE_{bal} exceeds it in all three (+0.162, +0.067, +0.123).

11. Discussion and limitations

What carries the shortcut decides the method. Across six real and controlled settings three regimes emerge. (1) *Distinct overlays that carry the shortcut in their pixels* (sonographer calipers, synthetic rulers, debris and tubes): masking removes an out-of-ROI copy but leaves an in-ROI copy intact, and U-MtE removes most of what masking leaves (thyroid +0.22, +0.20, +0.13 on three backbones; controlled sweeps +0.08 to +0.28).

(2) *Artifacts that resemble the pathology* (capsule debris versus erosion fibrin): any subspace estimated from artifact appearance removes disease evidence; disease protection makes erasure safe, and with image-level artifact labels masking plus last-layer retraining is the most robust option. (3) *Artifacts whose shortcut is carried by correlates* (dermoscopic hair co-varies with body site, age and sex; a chest drain comes with a treated, re-expanded lung): removing even every artifact patch with an oracle mask gains little (+0.015 for hair), no pixel- or insertion-based method can close the gap, and label-based group methods (DFR, balancing, JTT on hair) are preferable. The oracle-removal ceiling is a cheap diagnostic of the regime.

Why erasure works where masking and patch removal do not. In thyroid ultrasound, dropping every patch token that contains a caliper (oracle masks) before mean-pooling improves the reversed AUROC by only +0.047 over pooling all ROI tokens, whereas erasing the generic-overlay subspace from the class-token representation improves on masking by +0.22 (the two representations differ, so the comparison is indicative). Self-attention distributes the information about a salient overlay across the whole representation; an in-ROI artifact is therefore not contained in “its” pixels or patches, and a representation-level operation is needed to remove it.

Erasure, not augmentation. The same overlays used as training augmentation gain at most +0.03 over masking in every cohort; used to estimate and erase a subspace they gain up to +0.21 more. Augmentation teaches invariance to the *generated* overlays; erasure removes the directions along which *any* overlay moves the representation, which transfers to real artifacts the generator never produced.

Reading reversed-environment gains. A reversed-test gain can be manufactured by removing diagnosis-correlated signal together with the artifact: erasure fitted on unpaired image-level labels and post-hoc prevalence recalibration invert the correlated/reversed ordering. We report reversed, correlated and clean AUROC for every method and summarise robustness by the minimum of correlated and reversed AUROC.

Decision rule for practitioners. (1) Measure where artifacts sit relative to the ROI; out-of-ROI artifacts are removed by masking. (2) For in-ROI artifacts without any artifact annotation, use protected U-MtE; it matches oracle-mask LaMa inpainting where the artifact is a distinct overlay. (3) With image-level artifact labels, combine masking or U-MtE with group balancing or last-layer retraining, choosing among them on a group-aware validation split; prefer label-based reweighting alone when the shortcut is carried by correlates of the artifact, which the oracle-removal ceiling (a handful of annotated masks) reveals.

Limitations. (i) U-MtE is a representation-level intervention for frozen (foundation-model) features with linear heads, the setting in which it is evaluated throughout; end-to-end fine-tuning is a secondary check (the location law replicates; post-hoc erasure of a fine-tuned network’s features does not help, whereas erasing during fine-tuning works for ResNet-50 and ViT-S on thyroid), not a full sweep. (ii) Artifact masks

for thyroid calipers and capsule debris come from an audited rule-based detector (precision/recall 0.86–0.89) and a patch-token probe (IoU 0.62); errors act as label noise on A and on the overlap strata. (iii) Hair and ruler share one annotation channel in the ISIC 2019 archive. (iv) No patient or video identifiers are published for TN3K, MMOTU or SEE-AI; leakage groups use perceptual hashes and frame blocks, and a stricter embedding-based grouping leaves 11 of 12 primary estimates unchanged (Sec. 9.6). (v) Chest-drain labels for pneumothorax-negative radiographs come from a detector. (vi) The controlled chest-X-ray sweep did not show the sign reversal: the encoder ignored extra-pulmonary tubes. (vii) Reversed environments are constructed by subsampling; real deployment shifts are less extreme, and the clean results bound the cost without shift. (viii) The disease-protection step needs diagnosis labels of artifact-free images, and the balanced variants need image-level artifact labels. (ix) The linear-Gaussian account is fitted per model; it cannot anticipate a shortcut carried by correlates that the clean and correlated environments do not expose (chest drains).

12. Conclusion

ROI masking is not a neutral preprocessing step. Its benefit depends on where the artifact sits: it removes shortcuts that live outside the ROI and leaves—or amplifies—those that live inside it. This held in four diseases, three modalities, three frozen backbones, two fine-tuned architectures and on unaltered data, and it follows from a two-parameter model of signal and artifact strength. For in-ROI artifacts, erasing a subspace learned from generic synthetic overlays removes the shortcut when the artifact is a distinct overlay, needs no annotation of the artifact, and should be disease-protected; when the shortcut travels with correlates of the artifact, reweighting with image-level artifact labels is required. We recommend that shortcut audits report artifact location relative to the ROI and evaluate mitigations on shortcut-conflicting cases, alongside clean and correlated performance.

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